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ECONOMIC POLICY UNCERTAINTY AND FDI FLOWS: EVIDENCE FROM INDIA

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ABSTRACT

Foreign direct investment is crucial for the development of any economy. It serves as a method for establishing stable, long-lasting and direct links between economies. To enhance the inflow of investments in a country it is required that the host nation serves a stable economic environment to the investors which would also mean that a country must have low rate of Economic Policy Uncertainty.

Apart from other factors, economic policy uncertainty serves as one of the important determining factors for any investor, as certainty reduces the gap between projected and actual investment returns. Therefore, the governments must aim to decrease the amount of Uncertainty of Economic Policies in their country.

This research tries to investigate how economic policy uncertainty affects inflow of FDI in India over the period of 2003- 2022. The study was conducted using Autoregressive Distributed Lag (ARDL) method, which has shown that, the EPU has negative impact on the flow of foreign investments in India in the long run & is quite prominent as compared to the short run which is showing a contradictory result which could be due to the sample size of the study or the asymmetrical behaviour of economic indicators.

KEYWORDS: Economic Policy Uncertainty (EPU), Foreign direct Investment (FDI), India

INTRODUCTION

Foreign Direct Investment is regarded as the key driving force of economic development worldwide. The impact of which is seen in the growth of income and the creation of new jobs around the world.

In general, FDI has been defined as capital inflows from overseas that are used to increase the economy's capacity for production and are "usually preferred over other forms of external finance because they are non-debt creating, non-volatile and their returns depend on the performance of the projects financed by the investors. FDI also facilitates international trade and transfer of knowledge, skills and technology." (Kumari & Kumar, 2020)

There are some major influencing motives which acts as the reason for the flow of foreign investments into the host nations. Some of these factors are GDP, exchange rate, trade openness, inflation. Although these factors play a key role in attracting FDI, there is one other important factor which has gained importance in the recent times, which is economic policy uncertainty. Economic policy uncertainty can be described as the ambiguity which surrounds a particular policy of the government of the host country which affects the decision making of the foreign investors and hinders the flow of investments, as these policies can be altered at any time by that government.

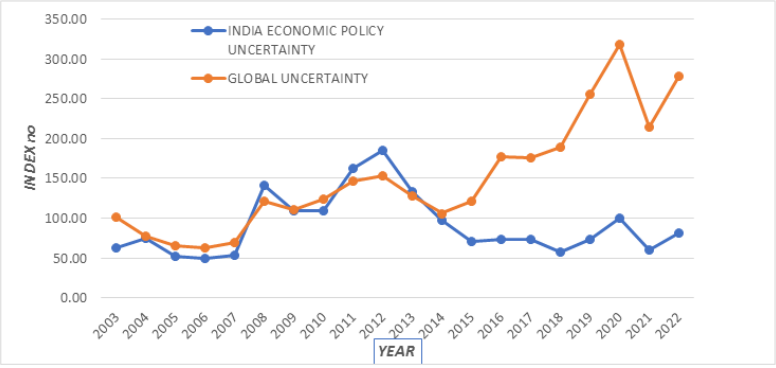
Many research studies have established that if the Economic policy uncertainty is high, it negatively impacts the inflow of investments as it is perceived as the indicator of limited protection by legal and political institutions of host country. The uncertainty of economic policies has deterring effect on the

inflow of investments as well as on production, stocks, trade credit and financial stability(Ogbonna et al., 2022).

The recent effect of uncertainty can be explained by the adverse impact caused by the pandemic on the economic functioning of the nations. As per the report of UNCTAD (2020), there has been a decline of 62 percent in the value and number of greenfield projects and a 72 per cent fall in cross-border mergers and acquisitions.

The uncertainty caused by the global financial and economic crises has also been seen to affect the economic situation of the nations, some of the examples are financial crises of 2008, European sovereign debt crises of 2010 and Arab springs of 2012 (Smith, 2021).

The aim of the current study is to comprehend how economic policy uncertainty affects the flow of foreign direct investments. This research is mainly focused on the Indian model which has shown to have decreased the economic policy uncertainties even in the times of economic crises and has managed to control its policies so that it does not affect the flow of foreign investments into the nation. The graph below shows the comparison between Global Uncertainty Index and Indian EPU Index.



***Source**: Policy uncertainty.com*

India's economic policy uncertainty has drastically reduced over last decade. It reached its highest point in late 2011 and early 2012, and it has since been dropping with occasional spikes. Until 2014, India's economic policy uncertainty moved in jointly with global uncertainty. However, it began to diverge in early 2015 and appears to have totally decoupled in 2018.Then, in the year 2020 there was a sudden spike in the Uncertainty index of the world and in India also. In the year 2021 it can be seen that the uncertainty index has dropped slightly but later in the year 2022 a little spike in the index was seen in both the indexes. But it can be clearly seen that as compared to the global uncertainty index the spike in Indian economic policy uncertainty index is low, which implies that the country has managed to control the economic policies thereby making the country a desirable place to invest. This research paper will further analyse the impact EPU has on the flow of FDI in India.

LITERATURE REVIEW

The present literature focuses on the determinants of foreign direct investments and their impact over the flow of investments. One such important determinant is Economic policy uncertainty which has gained importance in the recent times and its impact on FDI has been analysed in various research studies.(Haque et al., 2022) examined the effect of EPU on the inflow of FDI in nineteen higher income economies during the span of 2000-2021.It was observed that, exchange rate, growing levels of EPU and inflation has a negative impact on the flow of FDI into these nations. Research made by (Ogbonna et al., 2022) has determined via their research that the impact of global uncertainty on FDI made in Africa is enhanced by economic governance institutions. On the other hand,(Suleyman, 2020) found that regional and global uncertainty impact FDI flows in CEE Countries rather than domestic uncertainty.

Another important determinant of FDI is trade openness, which depicts the country's trade relations with the world and serves as one of the main factors in influencing the inflows of FDI. (Vuchkova, 2018) ,(Seetanah & Rojid, 2011) and (Boateng et al., 2015) have also established that openness of trade has positive effect in encouraging the flow of FDI. Research made by (Ibrahim & Gadir, 2015), (Ngo et al., 2020),(Koojaroenprasit, 2013), and (Ab, Muthiah, & Irfan, 2013) has established in their studies that FDI inflow is negatively affected by trade openness.

An additional important factor that affects the inflows of FDI is the rate of inflation.(khan & Rehman, 2017), (Haq, 2017)and (Geda & Yimer, 2018) has shown in their research that inflation has adverse influence on the flow of foreign investments. Another determinant called exchange rate was also analysed in various researches and is believed to have both negative and positive effects on the inflow of FDI. A study made by (Geda & Yimer, 2018) has explored the factor affecting FDI inflow in Nigeria during the year 1990 to 2017. The research reveals that the exchange rate positively affects the FDI inflow. Furthermore,(Gharaibeh, 2015), (Koojaroenprasit, 2013) and (Gupta, 2018) concludes that foreign direct investment is not positively affected by exchange rate.

OBJECTIVE

To study the impact of Economic Policy Uncertainty on the inflow of Foreign direct Investments in India.

METHODOLOGY

In this part, the variables used and the econometric techniques applied for analysis are explained. Although the study's main objective is to analyse the impact of EPU on flow of Foreign Direct Investments in India. But in order to obtain reliable findings, a number of additional explanatory factors have been included to the model. The data on FDI, Trade openness, consumer price index & Economic growth has been acquired from the World Bank database. The data about Real Effective Exchange Rate has been retrieved from Bank of International Settlements whereas Economic Policy Uncertainty from Policy Uncertainty Website. The research has used annual data from the year 2003 to 2022 in their study. The ARDL bound testing method is used to examine the long- and short-run effect.

MODEL SPECIFICATION

The impact of EPU on FDI inflow has been assessed in this study. The nature of the link among the relevant variables is ascertained using the functional form that follows.

$$FDI = f(EPU, TRADE, GDPG, CPI, REER) \quad (1)$$

$$LnFDI_t = \beta_0 + \beta_1 LnEPU + \beta_2 LnTRADE + \beta_3 GDPG + \beta_4 LnCPI + \beta_5 LnREER + \mathcal{E}_t \quad (2)$$

where FDI is the foreign direct investment inflow; EPU represents Economic Policy Uncertainty; *TRADE* indicates the degree of openness; CPI represents Consumer Price Index; GDPG demonstrates the economic growth of the country; and REER is real effective exchange rate in the Country. Note that variables are transformed into the natural logarithm except GDPG.

EMPIRICAL RESULTS AND DISCUSSION

ADF		***PP***			***KPSS***
Variables	***t-Statistic***	***P-Value***	***t-Statistic***	***P-Value***	***P-Value***
LnFDI	-3.432523	0.0253*	-3.079858	0.0454*	0.169984*
Δ LnFDI	-4.570986	0.0023*	-4.569380	0.0024*	
LnEPU	-2.125247	0.2377	-2.151423	0.2286	0.137689*
Δ LnEPU	-2.125247	0.0485*	-2.125247	0.0485*	

GDPG Δ GDPG	-4.001438 -4.698272	0.0070* 0.0020*	-3.989736 -12.61337	0.0072* 0.0000*	0.231394*
LnCPI Δ LnCPI	-1.522927 -3.405270	0.4995 0.0292*	-0.790588 -1.775969	0.7989 0.0379*	0.156853*
LnREER Δ LnREER	-2.405085 -4.562248	0.1535 0.0026*	-2.254999 -2.405085	0.1952 0.0278*	0.500000*
LnTRADE Δ LnTARDE	-4.562248 -3.713813	0.0026* 0.0134*	-2.937482 -3.721160	0.0596 0.0132*	0.253954*

Table 1. *Unit Root Test*

ADF, PP & KPSS TEST (Critical Value at 1% -0.739,5%-0.463,10%-0.347)

*Notes: Δ denotes first differences, Significant at *5% Level*

ADF, PP & KPSS unit root tests are employed in this research to examine the stationarity of these variables.

According to the findings in Table 1, LnEPU, LnTRADE, LnCPI, and LnREER are stable at first difference. The LnFDI and GDPG, on the other hand, are stationary at level. The ARDL model can be used in this research since the variables have a mixed order of integration.

Lag	LogL	LR	FPE	AIC	SC	HQ
0	-11.05150	NA	2.43e-07	1.794895	2.093139	1.845370
1	106.2111	148.1212*	5.81e-11*	-6.759060*	-4.671353*	-6.405737*

Table 2. *Lag Length Criterion*

**Indicate lag order selected by the criterion*

In this analysis, lag 1 is often used according to the AIC criterion and the f-statistic value obtained using lag 1 exceeds the upper bound value. However, in this study, lag 2 has been employed as, it was not possible to have a fitted model without it.

The ARDL technique divides estimations into two steps. The first step is to see whether there is a long-run equilibrium connection between the variables or not. For that, we have used the bound F-test.

F-statistics ***K*** ***Critical Bound Values***	5.419953 5	
Level of significance	***Lower Bound (I(0))***	***Upper Bound (I(1))***
10%	2.08	3

5%	2.39	3.38
1%	3.06	4.15

Table 3. *Results of bounds testing*

Table 3 illustrates the ARDL bound test's empirical findings. The study confirms a long-term effect between foreign direct investment inflow, EPU, openness to trade, consumer price index, real effective exchange rate & economic growth, as the F-value is higher than the upper bound value. In the second stage foreign direct investment's long- and short-term relation to independent variables are estimated.

Variable	***Coefficient***	***Std. Error***	***t-Statistic***	***Prob.***
LnEPU	-1.626164	0.307740	-5.284217	0.0132*
LnREER	-9.746886	2.647632	-3.681359	0.0347*
LnTRADE	1.595706	0.658837	2.422005	0.0940
GDPG	0.097187	0.032777	2.965088	0.0593
LnCPI	1.795300	0.366303	4.901138	0.0163*
C	35.23547	10.72206	3.286259	0.0462

Table 4. *Log-run estimated Coefficients (Dependant variable: LnFDI*),

*Significant at *5% Level*

Long-term estimates are presented in Table 4. The coefficients of the long-run relationship show that economic policy uncertainty, real effective exchange rate & consumer price index have a significant impact, whereas trade & economic growth is insignificant on the FDI Inflow.

The result showed that a 1 percent rise in the EPU reduces FDI inflow by 1.62 percent Furthermore, CPI positively influences FDI inflows, and the coefficient value of which is 1.795. The table also reveals a significant and long-term negative impact of the real effective exchange rate on the FDI. With an increase of 1 percent value of the effective exchange rate, the flow of FDI reduces by 9.75 percent in India.

Variable	***Coefficient***	***Std. Error***	***t-Statistic***	***Prob.***
D(LNEPU)	0.120996	0.099225	1.219411	0.3098
D(LNEPU(-1))	1.604155	0.234886	6.829506	0.0064*
D(LNREER)	-2.279786	0.562397	-4.053697	0.0270*
D(LNREER(-1))	5.223816	0.752320	6.943608	0.0061
D(LNTRADE)	1.693712	0.307618	5.505898	0.0118*
D(GDPG)	0.060537	0.012709	4.763337	0.0176*
D(GDPG(-1))	-0.119521	0.011245	-10.62850	0.0018
D(LNCPI)	31.90285	2.946285	10.82816	0.0017*
ECM	-1.720653	0.161282	-10.66860	0.0018*

Table 5. *Short-run estimated Coefficients*,

*Significant at *5% Level*

The short run results are summarised in table 5. The estimated coefficients of the short-run relationship show that EPU, real effective exchange rate, consumer price index, trade & economic growth poses a significant effect over the inflow of FDI. The coefficient value of EPU is 1.60. It means that a 1 percent surge in the level of EPU boosts the FDI inflow by 1.60 percent. while,1 percent surge in the REER reduces the inflow of FDI by 2.27 percent. Similarly, with an increase of 1 percent value of trade, economic growth & Consumer Price index the flow of FDI enhances by 1.69%, 0.06% and 31.90% respectively. Short run estimates are going against the prevalent theory either due

to the sample size of the study or because India's economic indicators are behaving in an asymmetrical manner. The ECM value presented above is −1.72 which denotes the speed of Long-run adjustments towards convergence.

ROBUSTNESS CHECK

	t-statistics	***P-value***
Normality test (Jarque-Bera)	1.064136	0.587389

Table 5. *Significant at *5% Level*

As the significance value is more than 5%, test employed in the study called Jarque-bera test supports that, the given model is normally distributed.

CONCLUSION

Economic Policy Uncertainty has become an important determining factor for analysing the economic situation of the country. Many researches have been done in the previous to establish the association between EPU and the flow of foreign direct investments. This paper has also examined the long and short run effect of EPU on FDI. For analysing this fact Autoregressive distributed lag (ARDL) method has been used which has shown that, the Economic policy uncertainty has a negative impact over FDI flows in India in the long run. It has also been established in the research that long run effect of EPU on foreign investments is quite prominent as relative to the short run which is showing a contradictory result which could be due to the sample size of the study or the asymmetrical behaviour of economic indicators. Although India has been quite successful in controlling the economic policy uncertainties in the past, but to further manage the economic policy uncertainty the policymakers must aim to create the policies more transparent and predictable so that the investors can make effective decision to invest without any fear of changing policies in India.

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